CNENGR202 Instrumentation and Control

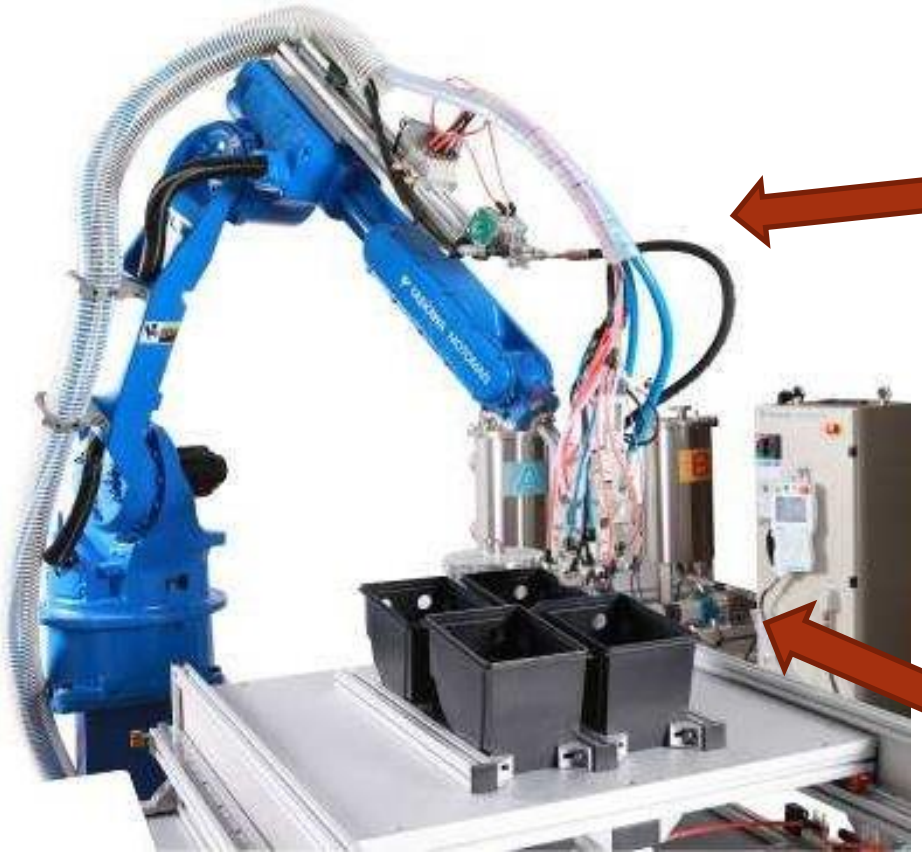
Part 1 L1 - Introduction

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Robot (I)

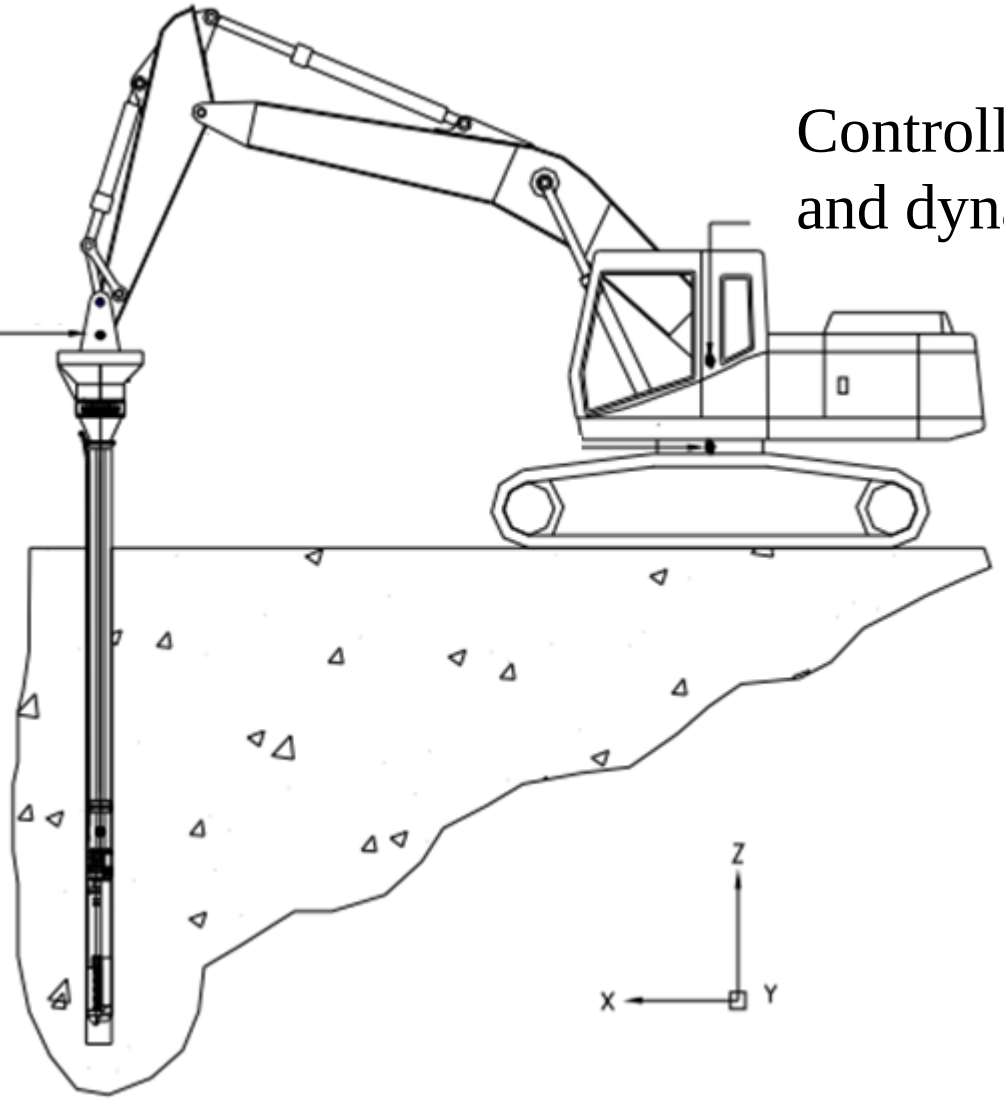


Instrumentation

Controller (Embedded system and dynamic control algorithm)

Ground Compaction

Instrumentation

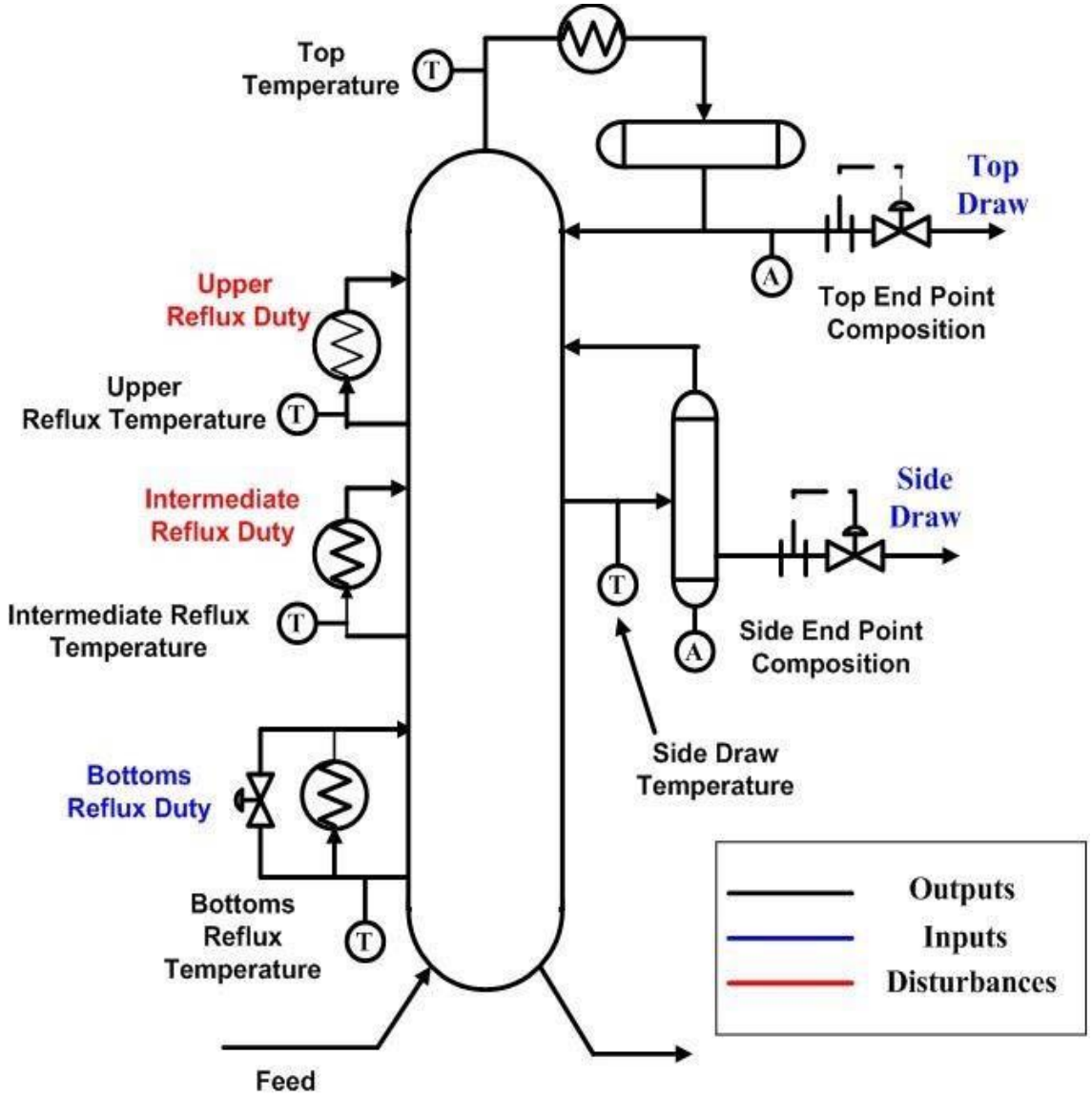


Controller (Embedded system and dynamic control algorithm)

Oil Fractionator

基准仿真用于比较各种多变量控制系统

- Benchmark simulation used to compare various multivariable control systems



Instrumentation & Control

Part 1. Dynamic Response of Systems Weeks 5-8

目的-在框图层面上发展对系统动力学和反馈控制的理解

Aim – to develop an understanding of system dynamics and feedback control at the block diagram level.

Part 2. Instrumentation 目标-培养选择和使用适当仪器进行反馈控制和数据记录的能力。 Weeks 9-12

Aim – to develop an ability to choose and use appropriate instrumentation for feedback control and data logging.

Part3. Embedded Systems Weeks 13-16

目标 - 考虑如何将内存、数字和模拟输入/输出设备与微处理器或微控制器进行接口。

Aim – to consider how to interface devices such as memory, digital and analogue I/O to a microprocessor or microcontroller.

Assessment

Assessment

- Exam based on wks 5–16 (80%)
- Coursework based on wks 5–16 (10%)
- Practical based on wks 5–16 (10%)

Part 1. Dynamic Response of Systems - 1

Part1

1 – Mathematical Models

2 – Time & Frequency Analysis

3 – Transfer Functions & Block Diagrams

4 – Automatic Control

Part 1. Dynamic Response of Systems - 2

Further Reading

Feedback control of dynamic systems

G.F. Franklin, J.D. Powell & A. Emami-Naeini

Global Edition (7th Ed.)

ISBN 10: 1-29-206890-6



Part 1. Dynamic Response of Systems - 3

Slides

- The slides for each Part will be placed on-line
- Take your own notes for case studies and for step by step maths examples etc.
- Each Part includes a set of self-assessment questions

Part 1. Dynamic Response of Systems - 4

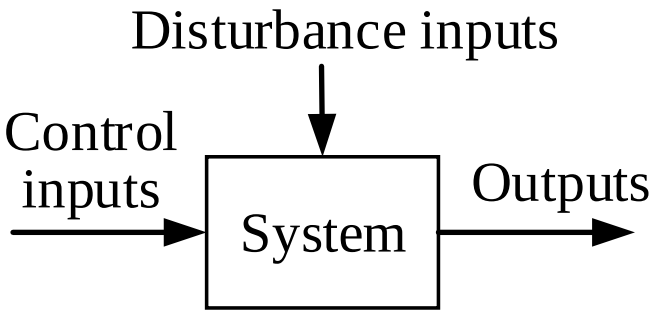
Timetable

- 2 lectures or experiments / week
- Lecture weeks 5, 7
- Matlab/Simulink laboratories weeks 6 and 8
- Experimental Report, for submission by end week 9

Part1. Dynamic Response of Systems - 5

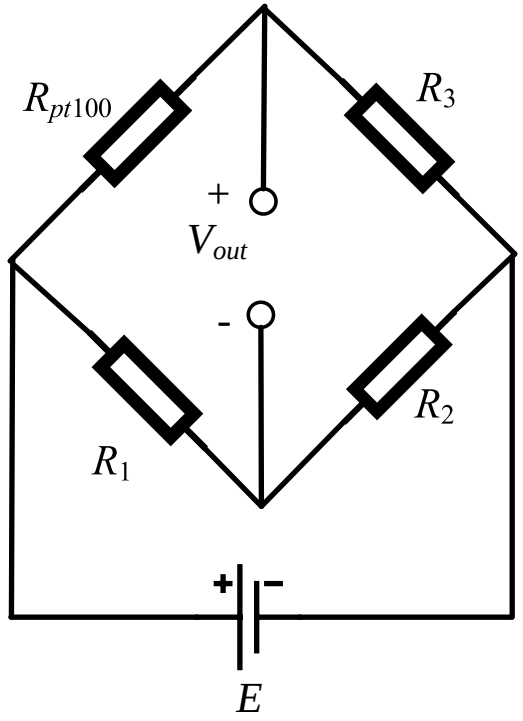
系统是由一组以某种方式连接在一起的组件组成的研究对象

- A **system** is an object of study consisting of a set of components connected in some manner.
- These components may be **physical**, chemical, biological or some combination of these.
- The system is typically defined by a number of **inputs** and **outputs**.
- Distinguish between **control inputs** and (uncontrollable) **disturbances inputs**.



Static Systems

- Most physical systems change with time.
- If the rates of change in a system are instantaneous, then the system is static.
- e.g. resistance network

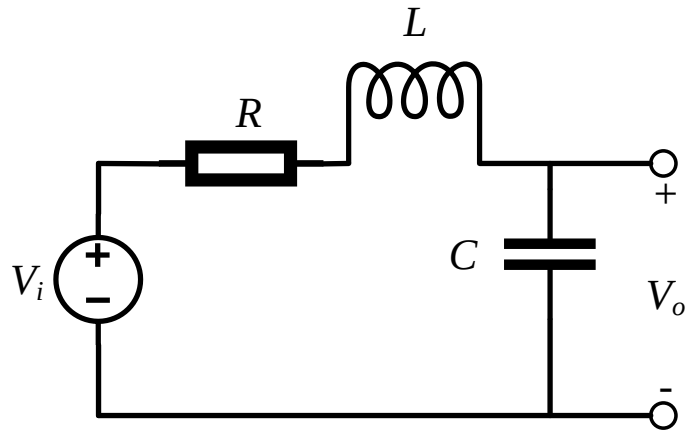


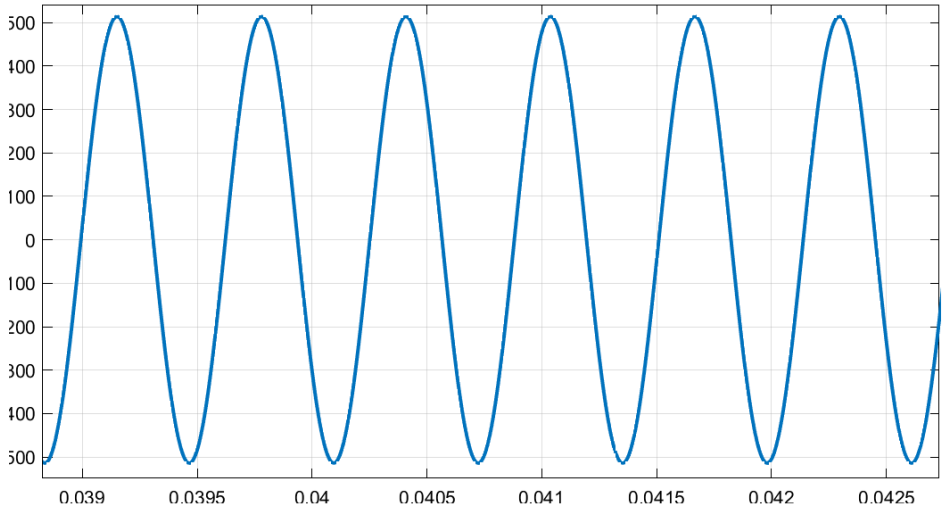
$$V_{out} = \left[\frac{R_3}{R_3 + R_{pt100}} - \frac{R_2}{R_1 + R_2} \right] \times E$$

Dynamic Systems - 1

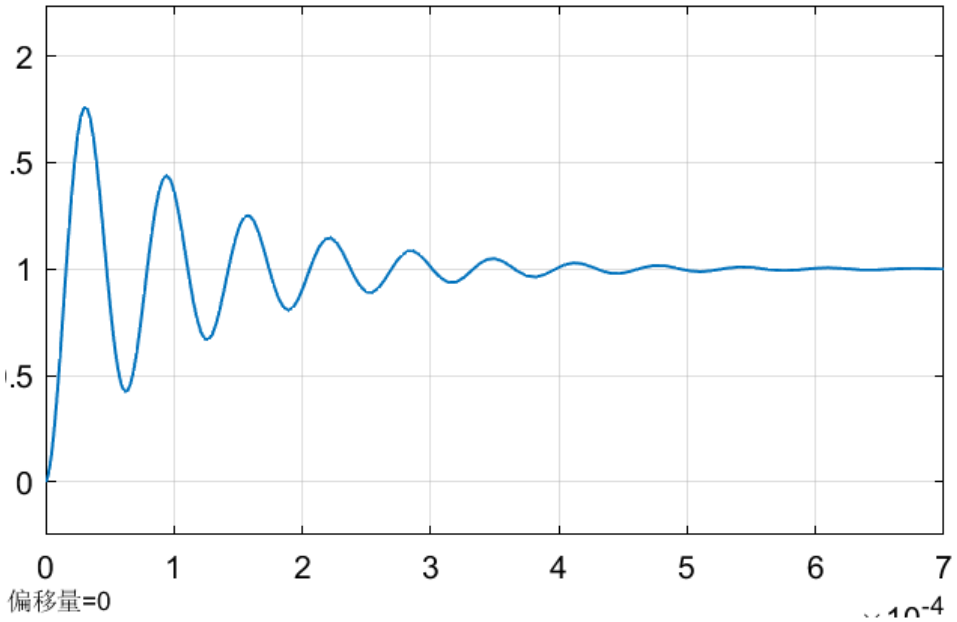
- If the rates of change in a system are not instantaneous, then the system is dynamic

- Example 1: an electrical circuit with capacitor and inductor





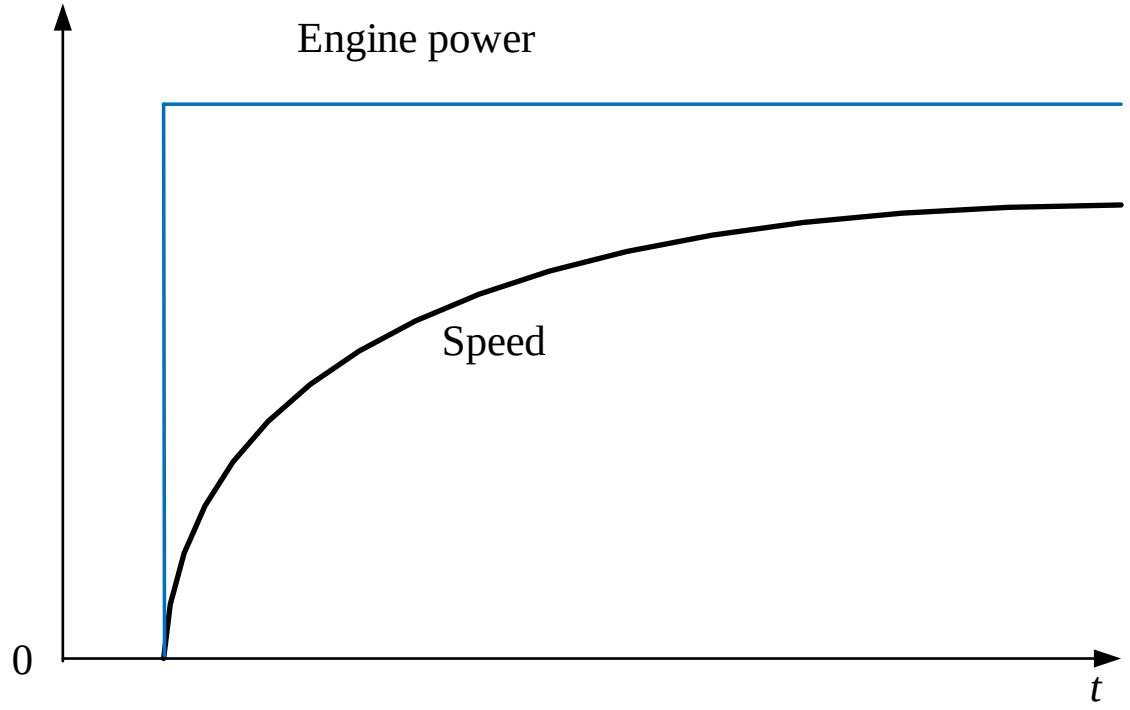
Ideal: no resistor



There is a 0.1Ω resistor in inductor

Dynamic Systems - 2

- Example 2: vehicle accelerates (gradual build up of speed)



Models for Dynamic Systems

动态系统通常表现为时变和非线性行为;**不幸的是，**我们所理解的大多数数学都是时不变的和线性的

- Dynamic systems usually exhibit time-varying and non-linear behaviour; unfortunately, most of the maths we understand is time-invariant and linear!

只要不违反某些假设，“实”系统通常可以用线性方程来近似

- The 'real' system can often be approximated by linear equations, as long as certain assumptions are not violated

- If the rates of change are continuous, then we obtain differential equations

- The examples of electronics, mechanical systems will show that the mathematical models for these often fall into two generalised types - 1st or 2nd order differential equations

Control Engineering

- Motorcar controlled by human, primarily direction and speed
- Wider issues such as safety, fuel levels & navigation will be considered by human
- Automatic control systems for a car include power steering, Anti-lock Braking System, cruise control system etc
- Automatic control systems for a power supply include output voltage/current, MOSFET switch, input voltage etc

Control Engineering Applications

- Applications in transport, power electronics, plant
- Industrial plants including nuclear, chemical processing, oil refining, manufacturing
- Robotics and intelligent machines
- Household and consumer devices
- Scale ranges from a single component (e.g. valve) to an entire system (e.g. power station)

- The controller might be a electrical device, a mechanical device, a micro-processor

Power Supply

- Automatic control of power supply, pulse width modulation, input voltage and output voltage



Automatic Control of Concrete ingredients

- Sand, stone and cement etc. are only possible with automatic control
- Weight of sand, stone and cement required with measurement
- To be able to understand, design and eventually control dynamic systems, mathematical models are needed

Question 1

Why use control system in plant, such as medicine, glass, power supply?